

Global Nickel Short-Term Outlook

January 2025



Lens Metals & Mining

A data analytics platform made for all enterprise users **to get immediate insights and foster collaboration**



Our global data coverage includes operating data on mines, smelters and refineries, supply chain and market analysis



Stay ahead of the market with timely updates on proprietary datasets to assess the metals and mining landscape



Opportunity screening made easy with geospatial context for your asset and market analysis using data from power, hydrogen, emissions, ports, railways and more



Commodities available:

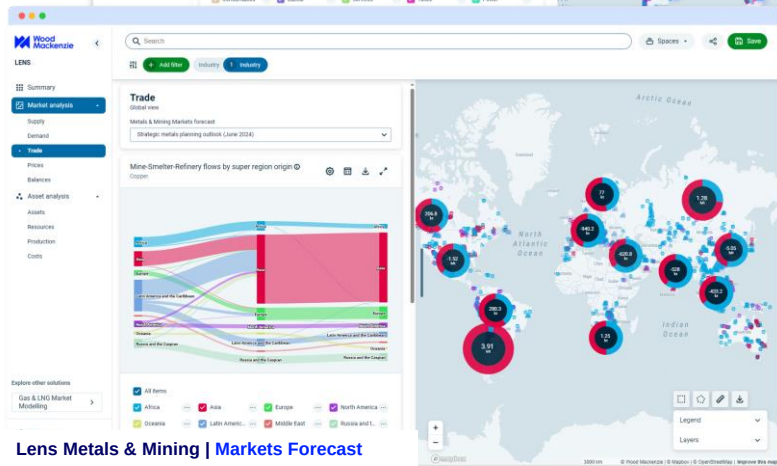
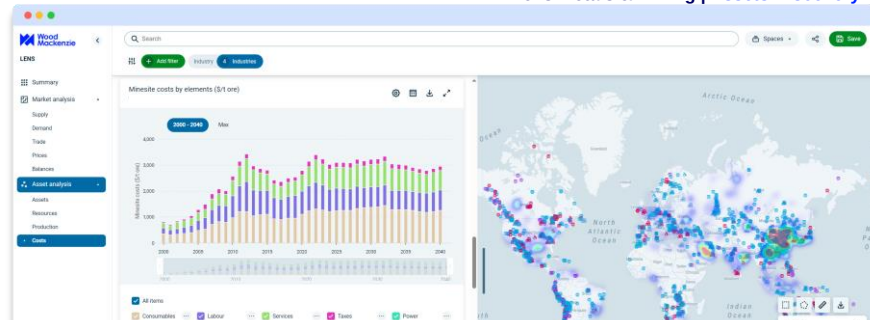
Copper Zinc Lead Nickel

More commodities coming by end of 2024



Scan QR code to learn more!

Lens Metals & Mining | **Assets Discovery**



Lens Metals & Mining | **Markets Forecast**

Executive summary

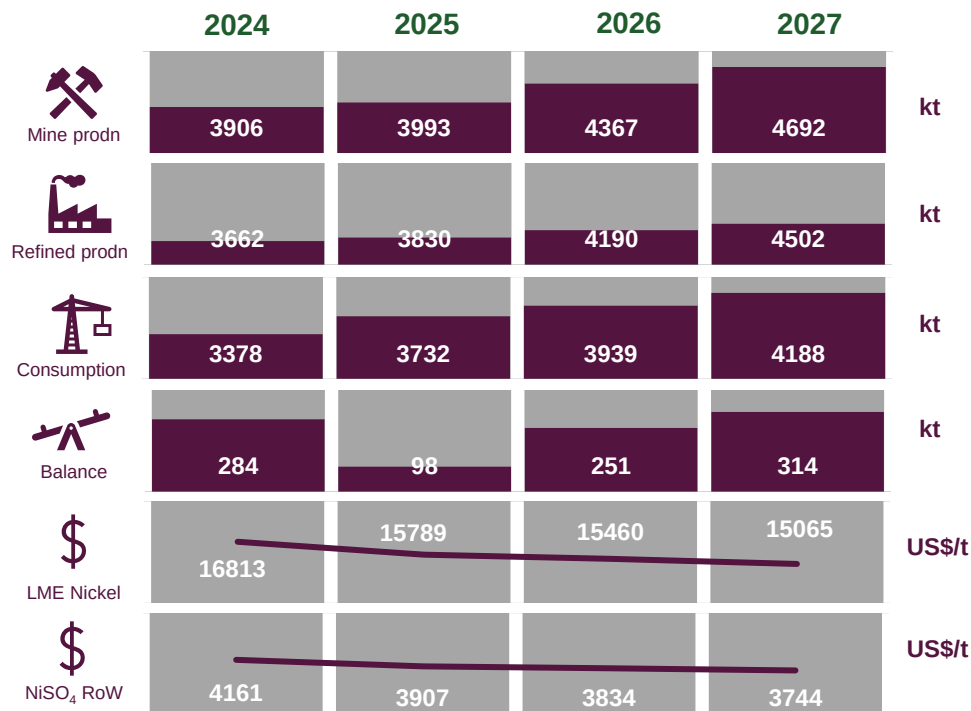
Nickel key forecasts

LME nickel prices defied reality once again for most of January as the bulls got carried away on trade press reports of Indonesia proposing to reduce permitted ore extraction rates for this year. Depending on the number you “believed”, the proposed cuts could have removed any surplus in the market. By 21 January, the increasing fervour took the three-month price past US\$16,000/t (US\$7.25/lb) for the first time since 13 December. And yet magically, just two days later, Indonesia confirmed that permitted mining quotas would be greater than last year. Prices have edged down ever since.

With nothing but the realities of the Chinese New Year holidays and slack demand firmly upon us, the market is wondering where to take prices next. At current levels of US\$15,250/t (US\$6.92/lb – cash), a return to January’s low of sub-US\$14,800/t (US\$6.71/lb) may not be too far away. Certainly, the risks are on the downside in the near term: rising stocks, tepid demand in both mainstay markets and surpluses in both Class I metal and sulphate.

And it could get worse. Our battery and precursor colleagues are assessing China’s nickel-based battery market for this year in light of ongoing gains by lithium phosphate and excessive precursor/CAM production rates, and the news is not good for nickel.

Any addition to our market surplus from that adjustment might be mitigated by one of the few potential “upsides” – further closures at western FeNi plants. But that is by no means guaranteed.



Source: Wood Mackenzie

Nickel balances by type

Class I – tight supply in the RoW in 2021-2022 supported nickel prices. The market moved into oversupply in 2024 and the cumulative surplus will continue to grow through to 2027.

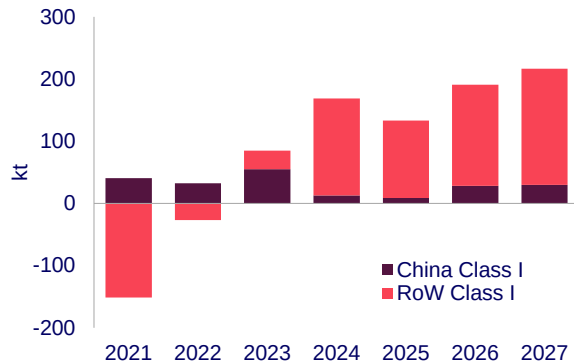
Class II – burgeoning Indonesian NPI production and sustained imports of western FeNi led to oversupply in China in 2022-2023. Closures at many western FeNi producers and stronger stainless production has tightened supply in 2024. Market will be even tighter in 2025-2026.

Nickel sulphate – a balanced market moved into oversupply in 2024, as Chinese battery demand has not kept pace with supply, itself boosted by expansion in supply of intermediates in Indonesia. Some of the excess has been further processed to Class I, which has gone to the exchanges, and this practise will continue in 2025.

	2021	2022	2023	2024	2025	2026	2027
Class I							
China Class I	40	32	55	13	9	28	30
RoW Class I	-151	-27	30	156	125	163	187
Total Class I	-111	5	85	169	133	191	217
Class II							
China Class II	0	163	202	-2	-64	-46	-47
RoW Class II	12	16	11	70	-31	0	0
Total Class II	12	179	213	68	-95	-46	-47
Nickel sulphate							
China sulphate	15	-5	13	56	50	70	61
RoW sulphate	-29	-27	-18	-9	9	36	83
Total sulphate	-14	-33	-5	47	59	105	144
Total nickel	-113	152	293	284	98	251	314

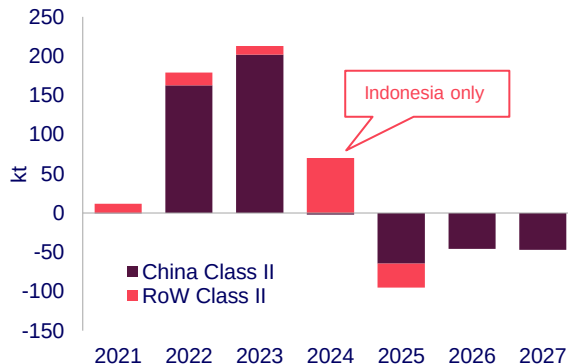
Source: Wood Mackenzie

Class I nickel balance



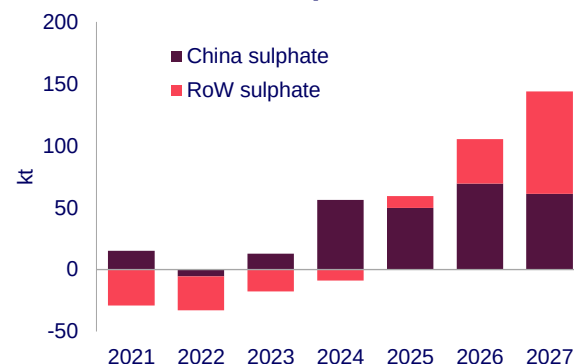
Source: Wood Mackenzie

Class II nickel balance



Source: Wood Mackenzie

Nickel sulphate balance



Source: Wood Mackenzie

Prices

Nickel heads back towards a floor

Indonesian mine permit strategy generated excitement and then disappointment

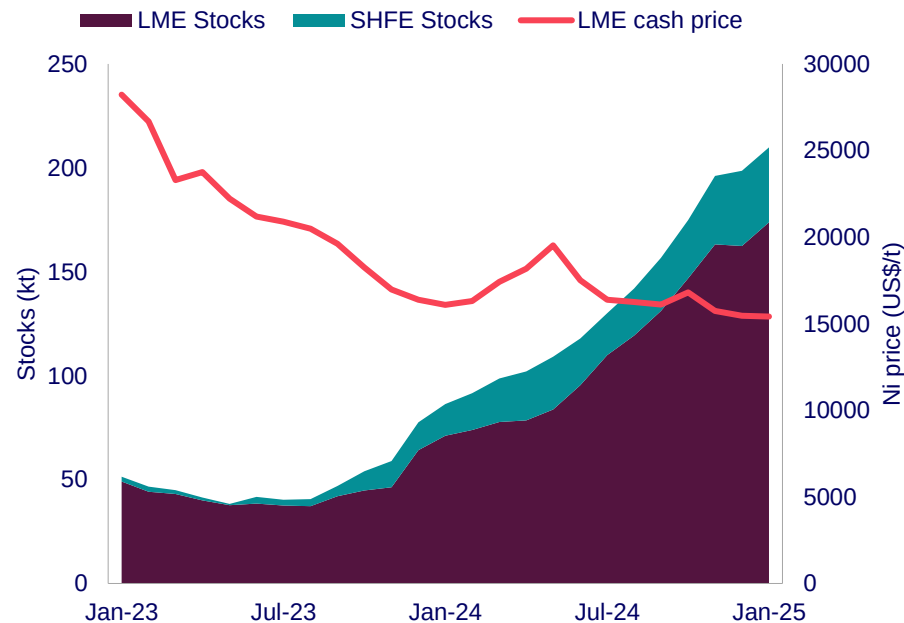
After hitting a low of US\$14,767/t (US\$6.70/lb, cash), at the close on 3 January, nickel prices proceeded to test the doubters by rallying steadily to an interim high above US\$15,800/t (US\$7.17/lb) between 17-21 January, with the three-month price hitting US\$16,000/t.

With the market flush with nickel, whether it be Class I, Class II or sulphate, the suspension at Ramu due to a failure in one of the acid plants was no doubt viewed as positive for prices. However, with persistent comments from the Indonesian government suggesting they would reduce nickel mining quotas in 2025, speculation on how large the cut would be was the key factor pushing nickel prices upward. We too believed there would be a reduction, although not to the low level that some had cited, so we too were caught out when it was confirmed on 23 January, that mining quotas in 2025 would increase to 298 Mt, from last year's 272 Mt.

Not surprisingly, nickel prices retreated, settling at US\$15,500/t on 27 January, before falling to US\$15,235/t (US\$6.91/lb) the next day, where they remain as we write.

The return of sub-US\$15,000/t prices looks imminent once more, particularly if LME stocks continue to rise as they have since September 2023. An average of 8 kt a month has arrived in LME warehouses since then, principally those in Asia, with a further 10.6 kt warranted in January, of which 6.5 kt went to Singapore and Kaohsiung. LME nickel inventory now stands at 173 kt which, together with SHFE, unchanged this month at 36 kt, leaves total exchange nickel stock at 209 kt.

Exchange nickel stocks and prices



Source: LME; SHFE; Wood Mackenzie

Exchange stocks will continue to rise in 2025

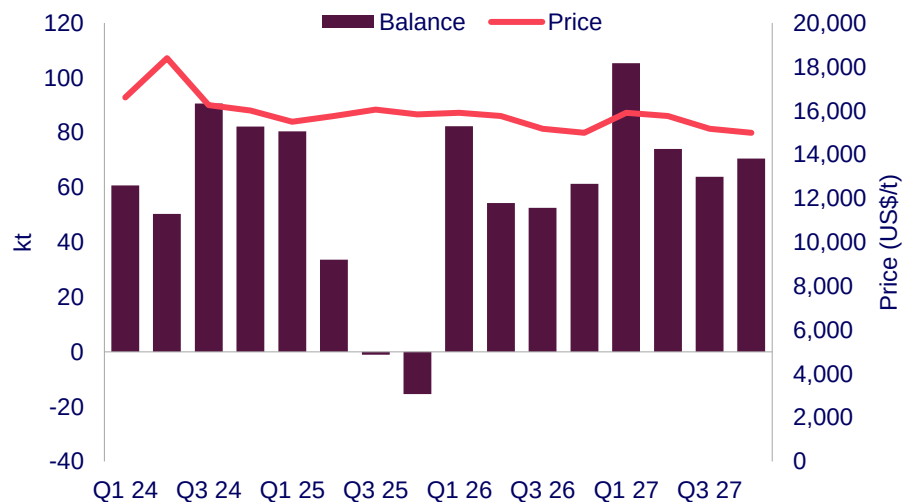
China and Indonesia are now producing Class I metal for stock

Realistically, this can only go higher given the outlook for Class I nickel production in China and Indonesia, and how this translates into a global Class I balance (slide #4). The shortage of Class I in the “west” prior to 2023 has been replaced by a surge in oversupply last year of around 150 kt, as the wave of new producers in China and Indonesia sent their finished product to the LME. There is no surge in demand specifically for Class I to match this supply – China and Indonesia are producing metal for stock. With nickel sulphate also in oversupply, more of the same is anticipated in 2025-2027, which should keep downward pressure on nickel prices.

As reviewed in our insight “Nickel: Five things to look for in 2025”, this pattern may change for two reasons. Firstly, there is some expectation among traders keen to try their hand on Chinese exchanges, that nickel trading in China may soon become more “internationalised”, either through adjustments to the existing SHFE contract or the introduction of new contracts on the Shanghai International Energy Exchange or through the Qualified Foreign Institutional Investor mechanism. At the very least this would modify the flow of Class I to the west, as would the LME’s intent to open a warehouse in Hong Kong this year. Equally, a new nickel sulphate contract was launched by Abaxx, in Singapore, on 10 January, which may also affect the volumes of Class I available for the exchanges.

So, what does the year ahead hold for nickel prices? On a positive note, we forecast 10% growth in global nickel consumption in 2025, a figure that ordinarily would push

Quarterly outlook for total nickel market balances and price



Source: Wood Mackenzie

nickel prices higher. Much of this is in China where stainless and batteries continue to expand, the latter recovering strongly on the assumption of a better economy. We make similar economic assumptions for other regions too, such as Europe, where strike-led closures last year mean that solid percentage growth in stainless is almost inevitable this year.

Global surplus to keep prices below US\$15,500/t through 2027

Downside risks could make things worse, although there are some positives to consider as well

All told, recovering economies return the global nickel market to balance in the second half of 2025, and on that basis, we currently see some better prices from around mid-year. But the improvement is not enough to move this year's average nickel price of US\$15,790/t (US\$7.16/lb) above that of 2024 (US\$16,813/t; US\$7.63/lb).

Naturally, there is downside risk. Having now started the year, to be talking of better economies ahead could be seen as optimistic. A weak January in China merely reflects the late-month holiday. February should be better, and that at least will feel good, but the strength of February will be the guide for the rest of the year.

However, in the Chinese battery segment, that pCAM production dropped below even last year's January figure is a red flag which, unfortunately, means our battery team will be reducing the forecast for Chinese nickel-in-pCAM for 2025 in February. We also assume that Xiamen Xiangyu will rectify the financial problems of Jiangsu Delong so that Delong can ramp up production at its Indonesian stainless plant – Obsidian. If this does not happen – and January guidance had no sign of this – then we will be reducing demand further. We are also concerned that industrial action may again disrupt stainless production in Europe, especially as the market there still does not appear to be improving. All of these will counter the current forecast of a tighter nickel market in H2.

The message from this is that our forecast of ongoing oversupply may get even worse. We forecast annual average prices falling to US\$15,460/t (US\$7.00/lb) in 2026 and US\$15,065/t (US\$6.85/lb) in 2027, guided by the 70th percentile on the cost curve, typically where prices head to in such times of excess. One hope is that after highlighting that the preferred price range for nickel would be US\$15,000-18,000/t (US\$6.80-8.15/lb), the Indonesian government would find the continued price decline unacceptable. Having implied that ore mining quotas would be cut this year, only to not follow through on that promise, that nickel prices continue to fall may persuade the authorities to make deep cuts next time, thereby helping to re-balance the market and turn prices around.

Added to that, Doniambo remains at risk. Financial support was to last until the end of 2024 so the market waits to see what happens next. There is also a suggestion by some analysts in the press that Cerro Matoso is “loss-making”. Our cost research shows that at current nickel prices and discounts for FeNi (average 17% in 2024), the asset sits in the upper end of the fourth quartile of the cost curve. Should this be enough to suspend Cerro then between these two operations around 70 ktpa nickel is at risk. That too would help to re-balance the market.

Nickel sulphate supply/demand and price

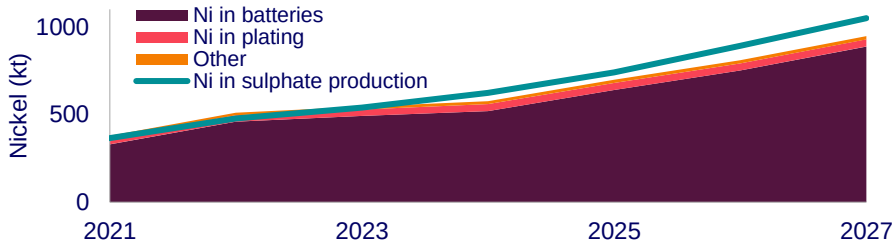
Premiums outside China are stable. Chinese sulphate continues to be discounted to the LME nickel price

China: The Chinese domestic nickel sulphate discount averaged US\$1,315/t nickel in 2024. We forecast this to narrow slightly in 2025 to US\$1,228/t before widening again in 2026 and 2027, to around US\$1,500/t in real terms.

RoW: Sulphate prices outside China averaged a premium of around US\$1,930/t nickel in 2024. This is expected to fall to around US\$1,800/t in real terms over the forecast period.

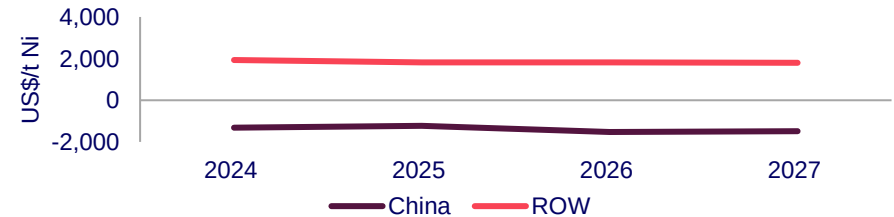
Indonesia: Chinese nickel sulphate imports from Indonesia were discounted more heavily than Chinese domestic material. Harita, the sole producer of sulphate in Indonesia, stated its sulphate sales price for nine months of 2024 as US\$14,344/t nickel – an average discount of US\$2,740/t.

Global NiSO₄ supply and demand



Source : Wood Mackenzie

Nickel sulphate premium/discount



Source : the Argus Media group, Global Trade Tracker, Wood Mackenzie

Demand

Changes in forecast nickel demand this month

Moderate revisions in stainless in 2026 and 2027, but net impact on nickel consumption is small

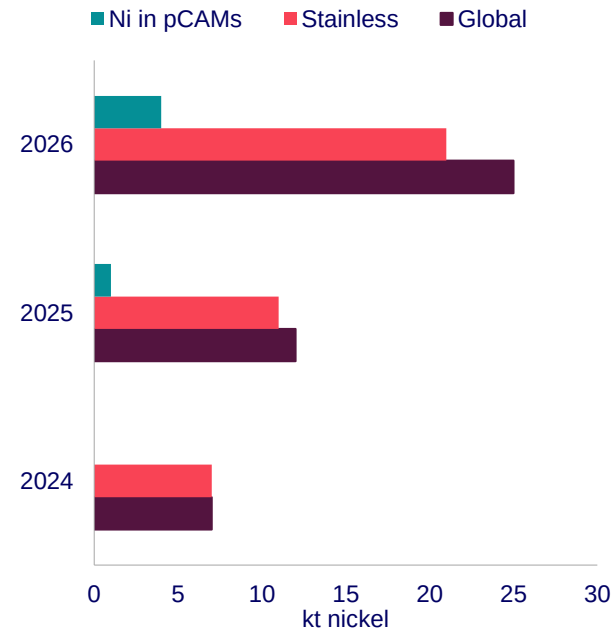
The main changes to the outlook in January are for stainless steel. The net changes to nickel consumption are small, increasing from 7 kt in 2024 to 25 kt in 2026.

Global output of stainless melt is revised up by 42 kt in 2024, tied to the latest data updates received for China, Indonesia, India, Europe and the US. The largest change was a cut in India of 204 kt.

A strong December in China for stainless production led us to extend that trend into 2025 and 2026, raising output by 450 kt and 640 kt, respectively. A slower ramp up at Jiangsu Delong's Obsidian plant in Indonesia warranted a cut of 140 kt in 2025.

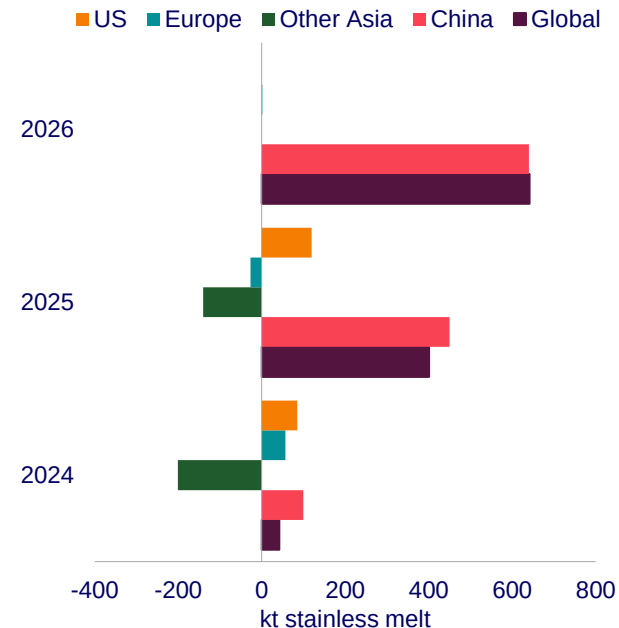
In pCAMs for batteries, nickel consumption was revised for Indonesia only to reflect a faster ramp up at new pCAM capacity.

Main changes in nickel consumption



Source: Wood Mackenzie

Main changes in stainless melt output



Source: Wood Mackenzie

Short-term outlook for stainless steel – global growth driven mostly by China

2025 is better, assuming global economies improve, but downside risks are starting to show

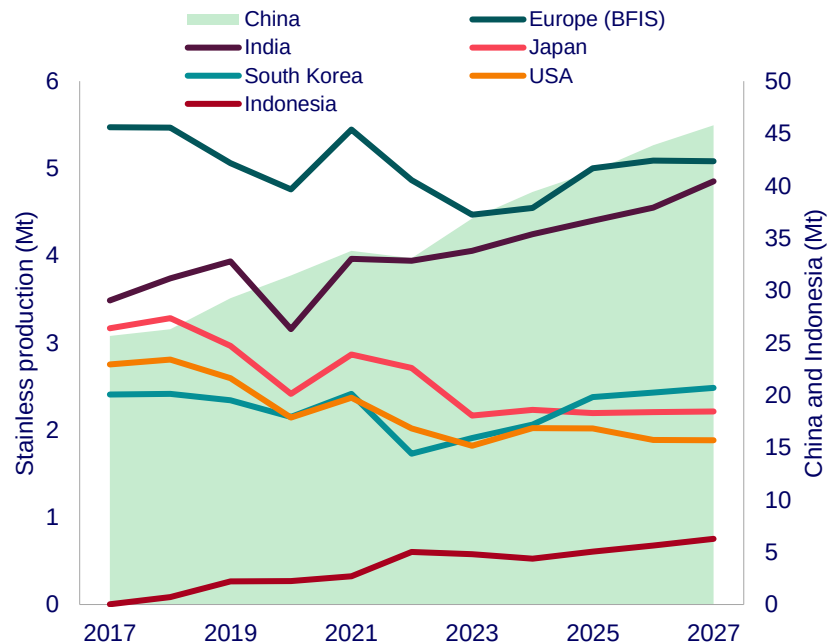
Global stainless production increased by an estimated 6.5% in 2024, to 62.9 Mt, and is forecast to increase at around 5% a year through 2025 and 2026, before slowing to 3.6% in 2027, reaching 72.5 Mt in that year. Growth is primarily driven by China, which accounts for more than 60% of global output. However, we also forecast improvements in Europe (better economy and no strikes) and Indonesia (through a rejuvenated Delong and new capacity from Jindal). Delong's ramp up may prove tardy, however.

European growth of 8% in 2025, to 6.8 Mt, may appear optimistic right now. However, this assumes the regional economy will pick up and that both Acerinox and Outokumpu produce uninterrupted, although the latter may already be affected by renewed industrial action. Regionally, however, the outlook for production remaining less than 7 Mtpa is well below the pre-pandemic rates of 7.4-7.5.

Some European mills have imported Indonesian NPI this year and this practise will continue in 2025, for as long as scrap supply remains tight. Therefore, European primary nickel consumption will rise more steeply in 2025 than the increase in austenitic stainless alone may indicate.

Global primary nickel consumption in stainless will increase from 2.18 Mt in 2024, to 2.39 Mt in 2025 and then to 2.47 Mt in 2026, net growth of 13% over two years.

Stainless steel production outlook by country



Source: Wood Mackenzie

BFIS=Belgium, Finland, Italy, Spain

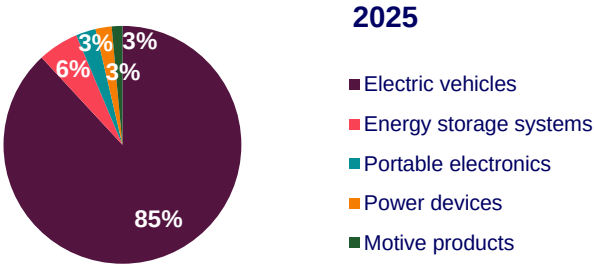
Short-term outlook for nickel in battery precursors (pCAMs)

Mounting risks to what has long been a positive outlook for nickel chemistries

The ongoing weakness in battery and battery materials markets continues. Global nickel consumption in pCAMs increased by an estimated 5.8% in 2024, to 520 kt, and will rise with a CAGR of 20% to 887 kt in 2026, most of this coming in China, albeit with some risk on the downside.

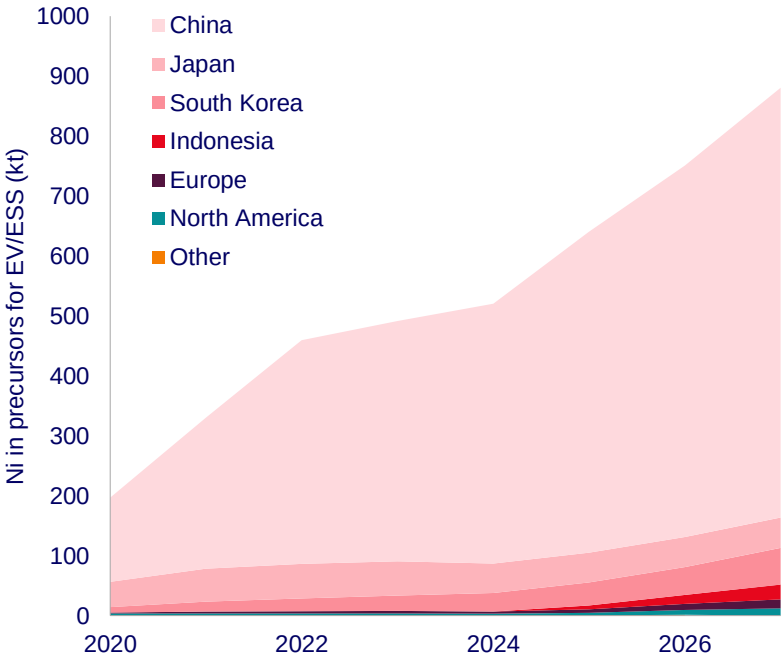
Japan and South Korea are the other two main pCAM producers. But while Japanese demand for nickel is steady at around 50 ktpa, new capacity in South Korea will boost its demand from 31 kt in 2024 to more than 60 kt in 2027. New Indonesian capacity will also start to ramp up this year, potentially reaching 25 kt of nickel consumption in 2027..

However, risks remain on the downside, the main threat outside weaker-than-expected EV and battery markets being the preference for LFP (lithium iron phosphate) chemistry, which contains no nickel and could account for 50-60% of the battery market in the future.



Source: Wood Mackenzie

Global nickel consumption in pCAMs



Source: Wood Mackenzie

Short-term outlook for global nickel consumption

Asia, especially China, is driving growth due to regional dominance in both stainless steel and batteries

Global nickel consumption increased by an estimated 6.8% in 2024, to 3.38 Mt, boosted by a substantial increase in demand in China, despite weakening stainless and battery markets. Battery markets, while better than in 2023, were under par and that has continued into January 2025.

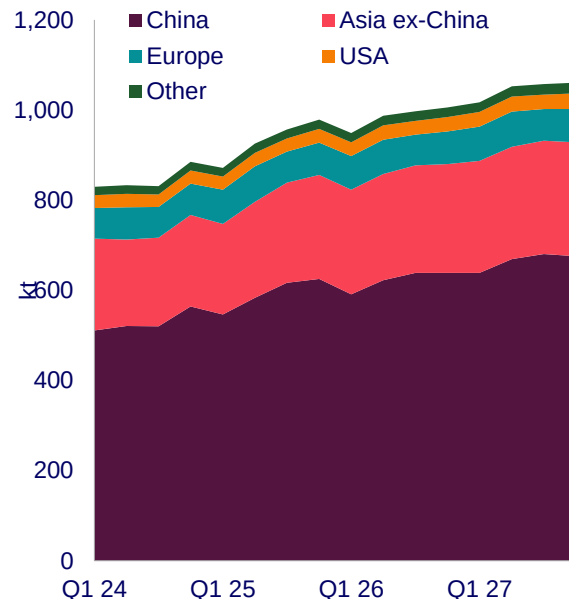
Elsewhere, increased stainless production supports stronger nickel demand in Indonesia and India, while use of Indonesian NPI in Europe has boosted primary nickel use there, and will do so again in 2025.

Growth of 10.5%, to 3.73 Mt in 2025, is mostly accounted for by China (helped by economic stimulus and a fall in stainless scrap use) and Indonesia, and also assumes recovery in other regional economies.

Growth will then slow to 5-6% a year through to 2027, with global consumption rising to 4.19 Mt in that year.

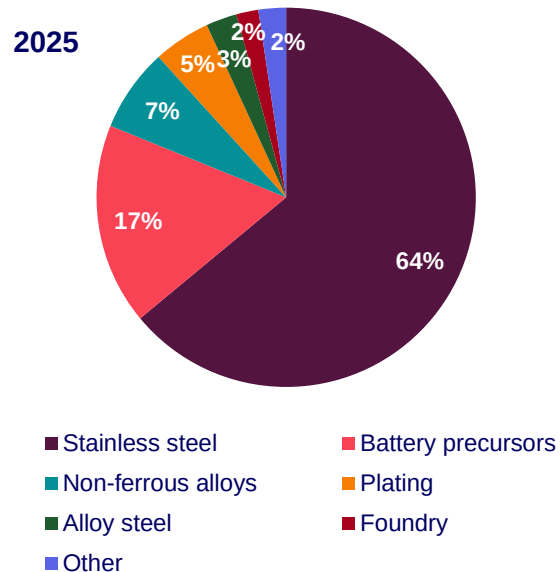
Stainless remains the principal consumer of nickel although the share of global demand going to battery pCAMs is growing, from 15% in 2024 to 21% in 2027.

Refined nickel consumption by region



Source: Wood Mackenzie

Refined nickel consumption by first use



Source: Wood Mackenzie

Chinese stainless production still rising despite off-colour demand

Demand is not keeping up with supply and stainless prices are still falling – to a four-year low in December

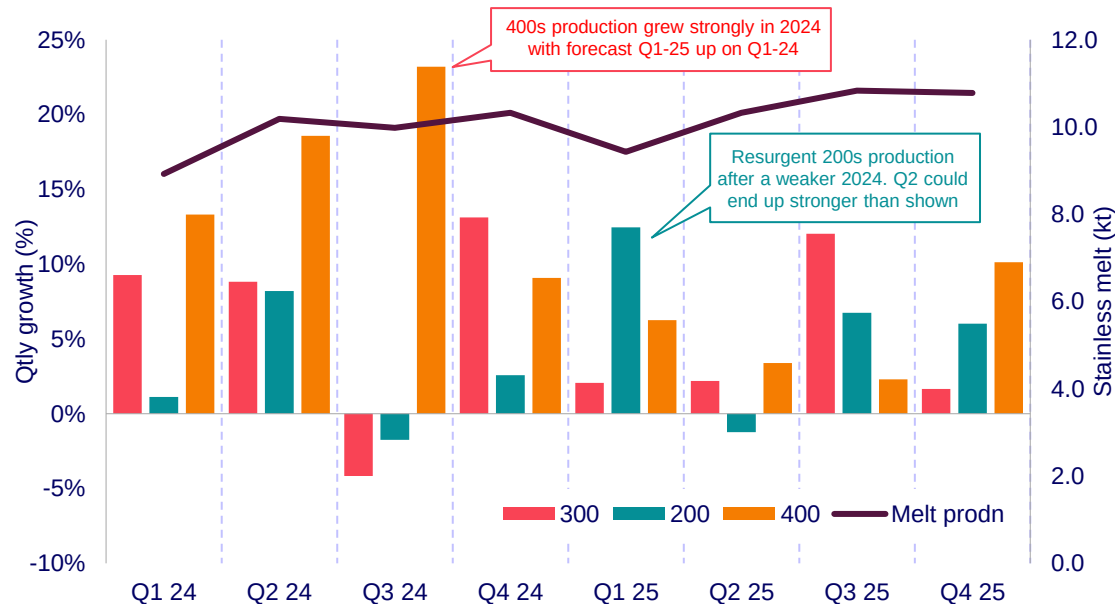
Economically, Chinese production ended 2024 with a strong December, with IP strengthening and exports also rising. Consumption and infrastructure also did well, and even the decline in property narrowed.

Exports increased in anticipation of new US tariffs on Chinese imports that would be introduced by the new US Trump administration. The export surge will, therefore, only last until those tariffs start, although with Trump so far refraining from a decision on this, strong exports will remain a little longer.

A further boost came in early January in the form of trade-in subsidies for appliances and autos, both of which will support near-term demand for stainless. Demand for appliances and metal goods, including kitchenware and cutlery, the main end-use segment for stainless in China, will be further supported by a slightly improved property market.

Metal goods is where most 200 series stainless, and more recently 400 series, is used in China, so the strong forecast growth in both grades in Q1 2025 (cf Q1 2024) is consistent with this scenario.

Changes in Chinese quarterly stainless production (year on year)



Source: Wood Mackenzie

Stronger Chinese stainless melting rates in late 2024 interrupted by Spring Festival

Forecast growth of 5% in 2025, to 41.4 Mt, as Tsingshan ramps up more new capacity and economy improves

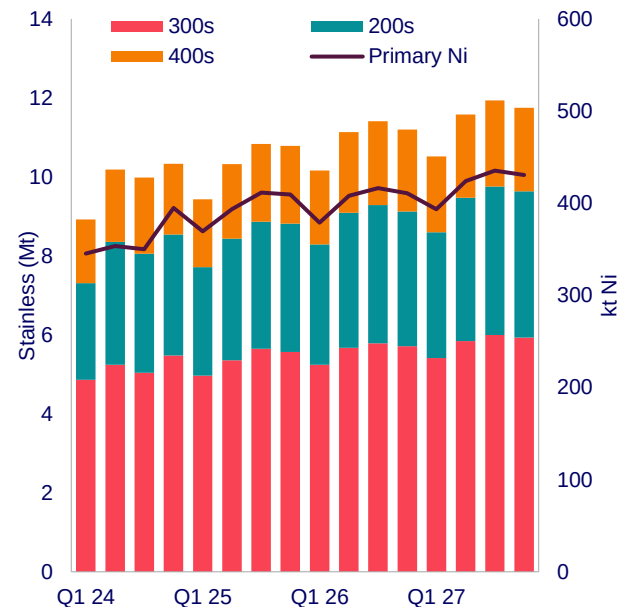
December stainless melt production in China was the strongest of last year, up 4% on November. This leaves Q4-2024 as the peak, at 5.47 Mt, and the year total at 39.4 Mt, of which more than half is 300s.

With the Spring Festival happening at the end January, it is not surprising that guidance for January was for a cut in output. Maintenance breaks of at least a week will cut production by 15-20% but this still leaves predicted national output more than 12 months ago.

Looking ahead, by extending recent production rates forward we expect 5% growth in 2025, to 41.4 Mt. This is supported by Tsingshan sustaining recent rates and filling new capacity that started up in late 2024, although there will likely be further reductions at other mills.

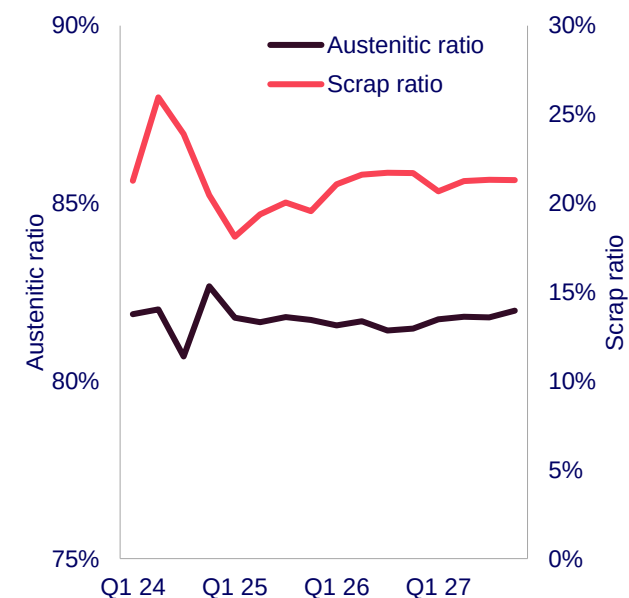
Optimism and higher NPI prices lie behind slightly better stainless prices in January. But, with ore quotas now increasing, the NPI price may retreat taking stainless prices with it, unless stainless demand is resurgent in February.

Chinese stainless production by grade



Source: Wood Mackenzie

Chinese scrap and austenitic ratios



Source: Wood Mackenzie

Indonesian stainless will increase when Jindal's new mill starts

But how much more stainless will Delong's Obsidian plant contribute between now and 2027?

The outlook for Indonesian stainless is for potential growth of 9% a year, to 6.5 Mt in 2027.

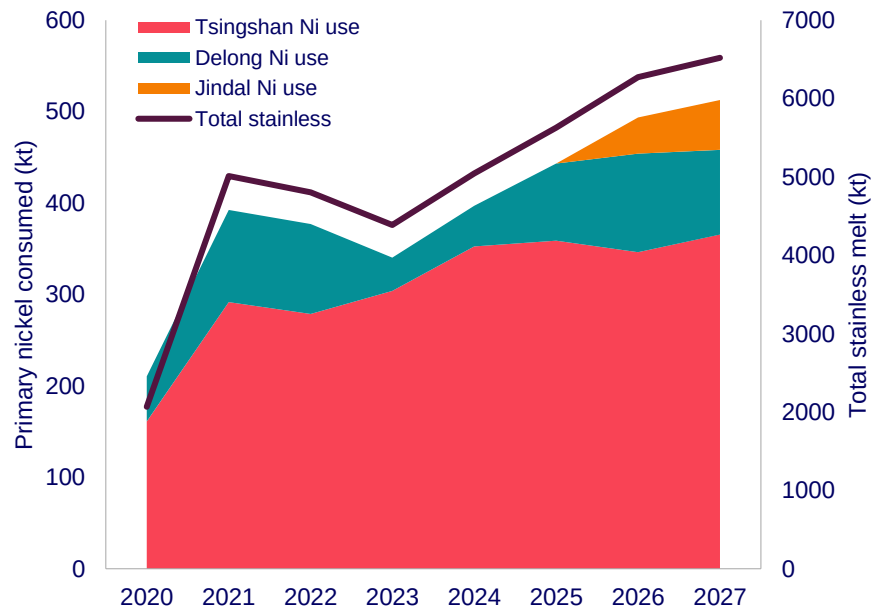
Tsingshan Morowali was producing at an annualised rate of 5 Mtpa at the end of last year and will not go higher without the addition of new capacity.

Delong Obsidian has 2.5 Mtpa of melting capacity but has produced less than 30 kt a month since August 2024, which is less than 15% utilisation. Anecdotal comments received this month indicate that the financial problems of Jiangsu Delong are still being resolved by Xiamen Xiangyu. We are optimistic that this will be achieved to enable Obsidian production rates to return to previous highs of around 1.5 Mtpa, perhaps from H2 2025.

Jindal's new 1.2 Mtpa stainless mill – **PT Glory Metal Indonesia** – which is now under construction, is scheduled to enter production in November 2025, potentially producing a forecast 700 kt by 2027.

Tsingshan, therefore, will remain the dominant stainless producer and primary nickel consumer in Indonesia over the forecast period. With austenitic scrap ratios less than 5%, total primary nickel uptake in stainless will exceed 500 ktpa in 2027.

Primary nickel consumption in Indonesian stainless

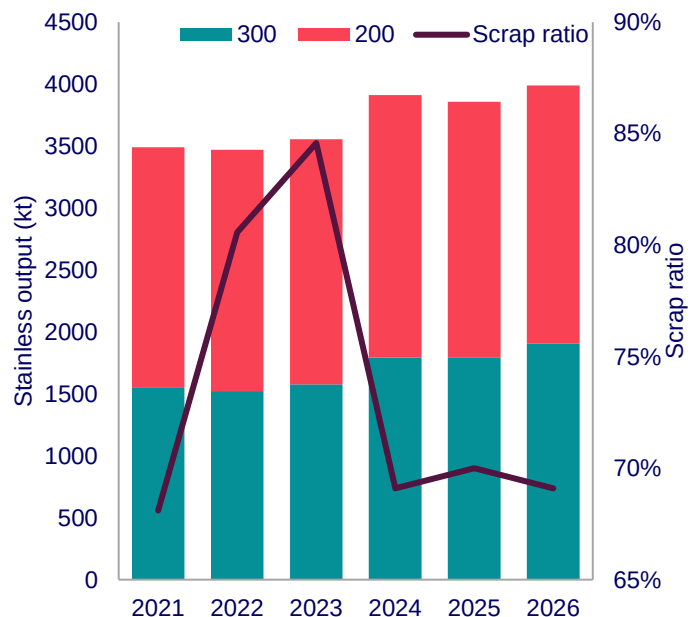


Source: Wood Mackenzie; ZLJ Steel

Indian stainless output revised down for 2024.

Domestic production capacity remains underutilised - ISSDA advocates for stainless policy

Indian stainless melt output by type, and scrap use



Source: ISSDA, Wood Mackenzie

Taking into account data from the Indian Stainless Steel Development Association (ISSDA) for the first nine months of last year, we have reduced our 2024 melt production estimate by 200 kt. This revision reflects the lower outputs in the second and third quarters, which fell by 8.4% and 5.7%, respectively, compared with the first quarter. Consequently, based on market trends and production outputs, we have set our 2024 estimate at 4.25 Mt.

Despite these adjustments, projections for 2025 and 2026 remain steady at 4.4 Mt and 4.55 Mt, reflecting an annual growth rate of 3.5% a year. The launch of Jindal's new melt shop in Indonesia later this year will have a bearing on national melt production as some intermediate stainless steel products (slab, hot-rolled coil) will be imported directly from the Indonesian facility. While this will provide for an increase in downstream stainless products output at Jindal (cold-rolled coil) it will not mean an increase in Jindal's domestic melt production. Elsewhere, we are aware of Shyam Metalics and Energy's plan to add 300 ktpa melt capacity at Sambalpur, Odisha, which might come onstream by 2027. While this is a new foray into flat products for Shyam, it will produce 200 and 400 series stainless via blast furnace and induction furnace combinations and will therefore consume no primary nickel.

ISSDA itself has been actively pressing the Iron & Steel Ministry for a separate policy for the stainless steel sector, which is currently included under carbon steel. Domestic producers face ongoing challenges due to the persistent dumping of low-cost Chinese products, resulting in loss of market share and underutilised melt shop capacity. India's total installed capacity stands at 7.5 Mtpa, with utilisation just below 60%. ISSDA's policy focus includes raw material securitisation, increasing demand, improving and increasing capacity. For our forecast period we foresee only moderate growth as India prioritises investments in the railway and infrastructure sector, which forms part of the automotive/rail/transport end-use sector and accounts for less than 10% of domestic stainless consumption.

European stainless forecast growth at risk

No meaningful signs of market improvement yet and strikes threaten Outokumpu Tornio again

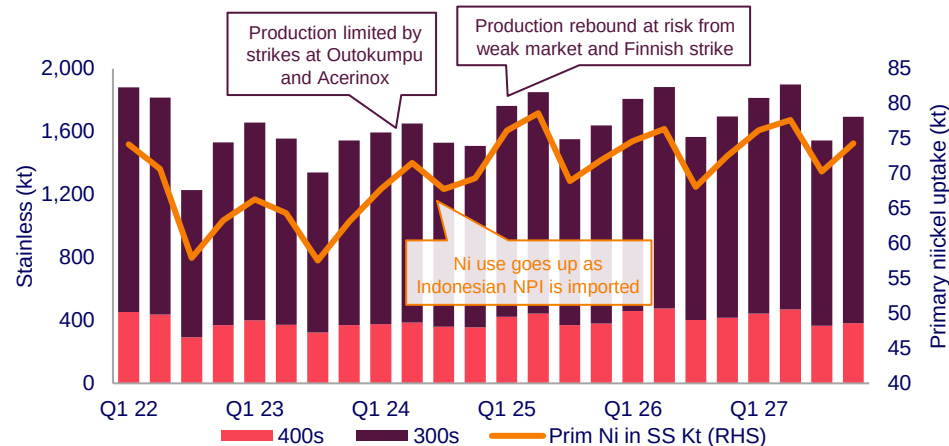
Unusually, on 7 January Aperam released a reminder to “...market participants of the standing guidance...” for Q4 2024 as outlined in its Q3 report. The re-stated expectation of improvement was perhaps aiming to reassure that the prior positive forecast for Q4 is still valid, despite the challenging market, and notwithstanding the fact that Q4 usually is better than Q3 for seasonal reasons. For the company's European stainless business, increased volumes are expected to be countered somewhat by downward price pressure.

If this is the result, it would be hardly surprising but what will be telling is what the volumes actually turn out to be, for Aperam and its European competitors. The same follows for Q1 which, together with Q2, is the strongest business period of the year. So, Aperam's confirmation in the above document that depressed volumes and prices in December will be important for Q1 signals that disappointment may lie ahead for this quarter.

Certainly, the European stainless market remains challenging, for both flat- and long-product makers. Prices are still falling according to Kallanish, with efforts by the mills to raise them being met with resistance from consumers. Inventories remain below average with no apparent interest in re-stocking from distributors, and lead times are short - some mills reportedly still trying to fill February's order book. “Normal” lead times would be two months.

Q1 already feels poor, so production rates may be down. And speculation circulates that Q2 will be likewise.

European stainless – growth in quarterly output, year-on-year



Source: Wood Mackenzie

Our forecast 8% growth in 2025 assumes regional economic improvement and is further boosted by the return to uninterrupted activities for two of the biggest producers who were temporarily closed in H1-2024 for strike-related reasons. Clearly this percentage figure is at risk, not least because one of the mills affected last year – Outokumpu - is facing a new round of industrial action in Finland. A six-day nationwide strike began on 27 January with follow up, week-long actions planned to start on 3 February and 10 February.

Aerospace demand strength undermined by Boeing safety issues in 2024

Ongoing recovery of aerospace is expected to continue and will support special (nickel) alloys demand

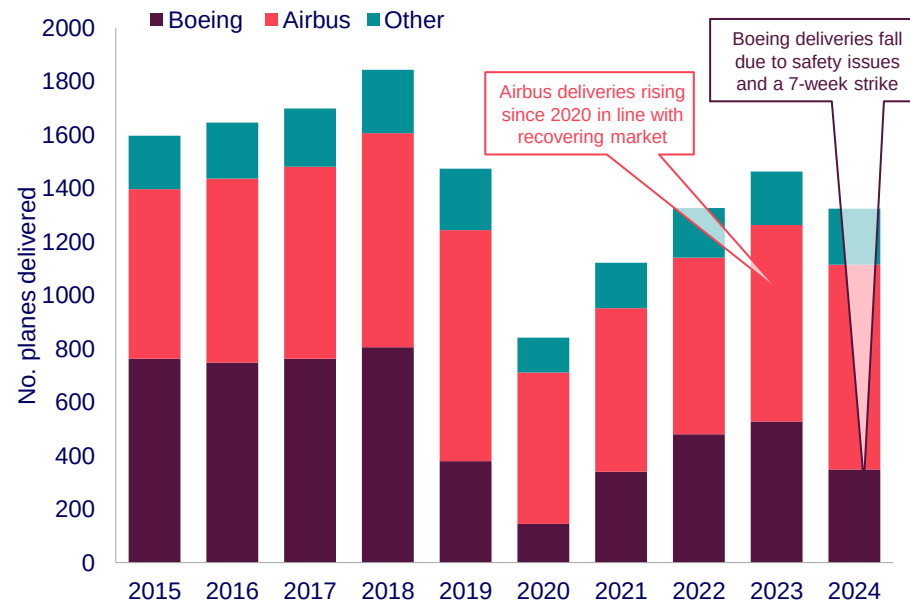
Full year aircraft orders and delivery data for the “big two” plane makers – Airbus and Boeing – show that the aerospace industry continues to grow after taking time to get going again after the pandemic. At least Airbus's results show that, deliveries increasing steadily from the low of 2020, to 735 in 2023 and 766 last year.

In contrast, Boeing has had issues: firstly, a couple of in-flight incidents led to groundings for safety checks and further FAA intervention; and secondly, a seven-week strike in the second half of the year. Between them this led to a 35% cut in deliveries, from 528 in 2023, still well below pre-pandemic levels, to 348 in 2024.

But Airbus has not been without its problems, with supply chain challenges, such as delayed deliveries of the CFM LEAP-1 engines, forcing the company's aircraft production target down from 800 to 770; so even that target was missed, albeit only just.

For nickel, which is consumed in stainless steel and, particularly, nickel alloys for aerospace applications, the current segment growth cycle clearly bodes well for demand. However, the long lead times between alloy melting and installation in, say, a jet engine, could mean that alloy already produced for use in 2024 will not now be utilised until this year, thereby moderating raw materials demand, at least in the near-term.

Aircraft deliveries

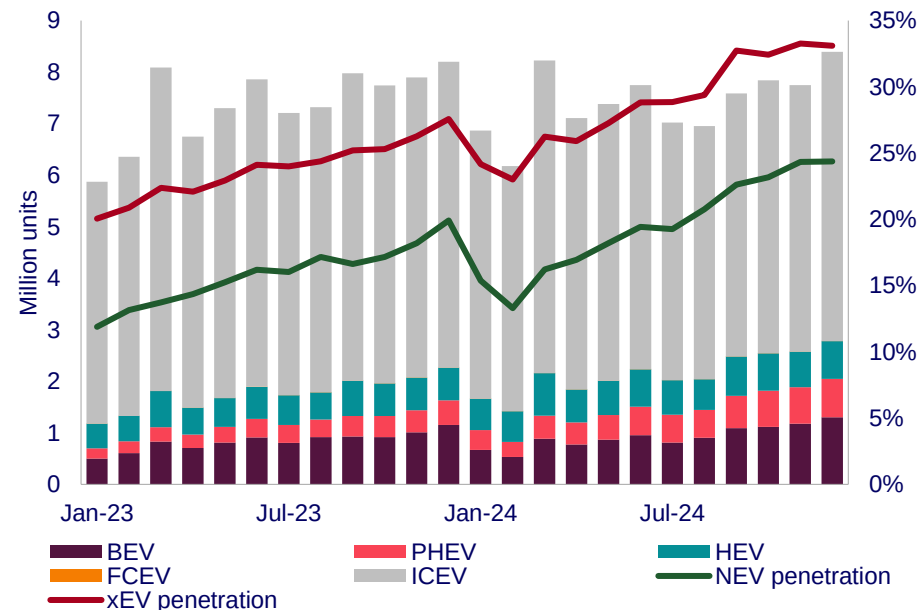


Source: Wood Mackenzie; Boeing; Airbus

Global vehicle sales hit 8.39 million in December

Full year sales reach 87.7 million, 1% higher than 2023

Global vehicle sales and PEV penetration



China: Concerns over the possible expiry of the government scrappage subsidy scheme drove wholesales in December to peak at 3.48 million, a 5% increase over November. Full year wholesales for passenger vehicles rose 6% year-on-year to 27.2 million. The move to EVs continues to drive the market towards domestic brands with all foreign OEMs except for Ford and Hyundai-Kia experiencing declines in wholesales in 2024. The newly-announced extension of the scrappage scheme will keep the momentum on demand in 2025.

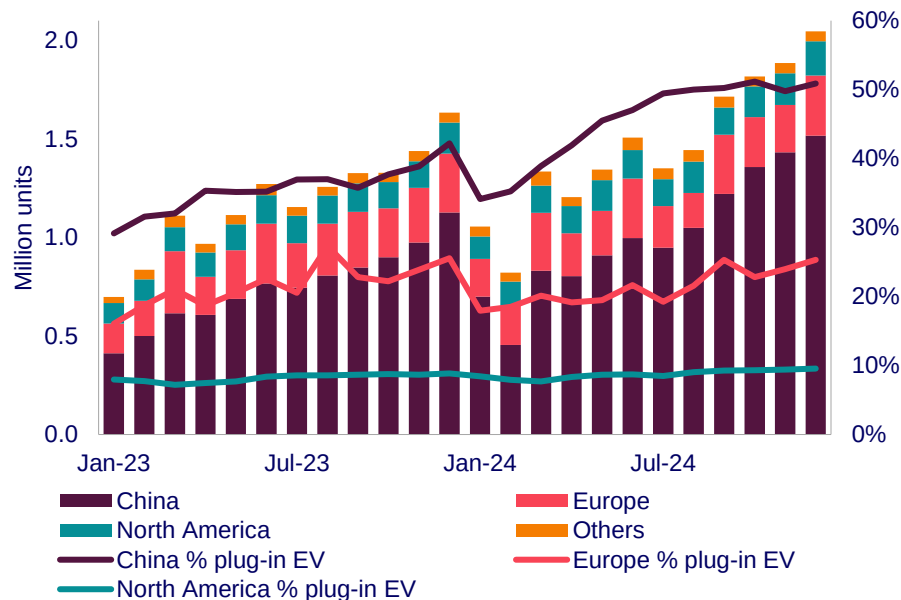
North America: Concerns over the possible impact of proposed tariffs on imports from Mexico and the proposed removal of the EV subsidy on auto prices supported growth in the US market. December sales hit 1.54 million, 9% up on November. Full year US sales rose 2%, to 16.4 million. The growth benefited most OEMs with only Stellantis, Tesla and Volvo experiencing year-on-year drops in demand. High incentives from OEMs and declining loan rates in early 2025 are expected to keep the momentum on demand.

Europe: December passenger sales grew to 1.2 million thanks to a significant bump in demand for EVs in the UK and Spain. Despite this, combined sales dropped 1%, to 15.47 million in 2024. Regional consumption levels continue to be dampened by high household savings rates. However, the parallel continuing growth in real labour income levels is expected to eventually raise consumption levels in 2025.

Global PEV sales exceed two million in December

Full year PEV sales hit 17.5 million, 24% more than in 2023

Sales of plug-in EVs by region



Source: Marklines, Wood Mackenzie, PEV = Battery-electric (BEV) + plug-in hybrid (PHEV)

China: OEM discounts, financing incentives and concerns over the possible expiry of the scrappage subsidy led December PEV wholesales to hit 1.51 million. Full-year sales rose 37% to 12.2 million, mostly via PHEVs. Full-year PEV penetration rose to 39%. Rising preference for domestic brands continues to affect the market with most foreign brands, including Tesla, experiencing a year-on-year drop in PEV sales.

With year-end OEM discounts and incentives set to be discontinued in January, PEV wholesales are expected to revise down marginally from December's peak.

North America: Concerns over the elimination of the PEV tax credit along with aggressive incentives from Tesla drove up US EV sales to 159,000 in December. The market achieved full-year sales of 1.58 million, an 8% rise from last year. Honda and GM's new mid-sized SUV BEVs contributed to the bulk of the growth. PEV penetration rose from 9.1% in 2023 to 9.6% in 2024. Continuing concerns over the possible elimination of the EV subsidy along with OEM aspirations to hit EV variable profitability via scale will drive sales through 2025.

Europe: Pressure on OEMs to comply with the UK's ZEV mandate, and pressure on consumers to avail Spain's EV subsidy prior to the program's expiry at the end of the year, drove December PEV sales up by 26% over November, to 304,000 units. Lack of subsidies, however, drove full-year sales down by 3%, to 2.89 million units in 2024. Spain's extension of subsidies and pressure on OEMs to comply with the EU's 2025 emission threshold will keep the momentum of PEV demand in H1 2025.

Battery raw materials analysis

Early signals from China imply suppressed nickel demand: holiday sentiment or underlying issues?

In January, nickel demand from the Chinese battery sector was weak. Because of the Chinese New Year holidays and a consequent slowdown in downstream activities, we had factored in a poor start to the year in our data. But building feed stocks for post-holiday business had already been completed earlier in the month and, with many cathode precursor producers having shifted into holiday mode, we saw reduced purchasing activity for upstream raw materials. Additionally, as downstream battery manufacturers typically experience longer resumption times after the break, the weak demand for nickel is expected to persist into early February.

Our current data show that, after weaker-than-expected performances during 2023 and 2024, when nickel demand in batteries in China still increased by 7.5-8%, we expect a strong double-digit growth rate exceeding 20% this year. This had nickel use in the battery segment rising to 530-540 kt from last year's 430-435 kt. However, market pressure from LFP batteries continues to suppress overall demand for nickel-based batteries. Given these factors, we anticipate that the demand for nickel from the battery sector will remain subdued in the near term. As a result, we will closely monitor the market after the holiday and consider the potential for further downward revisions to nickel demand in the coming months.

Morocco earns its place on the nickel-based battery map. This month CNGR Advanced Material commissioned part of its ternary precursor facility at El Jorf Lasfar, to the southwest of Casablanca. Of the US\$2 billion joint venture, CNGR Morocco New Energy will hold a 50.03% stake, with Moroccan investment fund, Al Mada, taking the balance via its fully-owned subsidiary, Next Generation Industries (NGI). Once the complex is completely built, it will have a battery recycling facility able to generate 30 ktpa of black mass, 120 ktpa of precursors and cathode active materials, and 60 ktpa of LFP materials.

We had expected a 2026-2027 start to the plant and currently show Morocco consuming about 4 kt of nickel resulting from this plant. However, trade data available to November last year appear to confirm the earlier start. Imports of nickel sulphate for 2024 reached 3.7 kt (containing 800 t nickel), compared with only 20 t in prior years. Almost all the 2024 imports came in from Nickel West's Kwinana sulphate refinery in Australia, which closed in July 2024. With cobalt sulphate imports also rising to 285 t (gross) from barely one tonne in prior years – almost all from CNGR's Finnish plant – it is clear that CNGR wants to tie in sourcing to be acceptable to the EU and US customer base, given Morocco's FTA status. Without Kwinana, though, it will be interesting to see where the long-term supply of nickel sulphate comes from.

Supply

Refined supply

Refined output to increase by 24% by 2027 compared with 2024

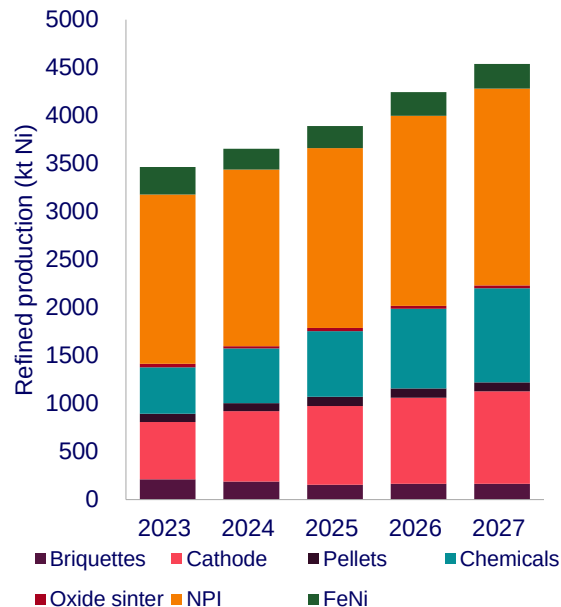
Class 1 output is forecast to increase by 21% by 2027, from 2024 levels, primarily from the refining of excess Chinese nickel sulphate to metal.

Chemicals output will increase by 73% over same period.

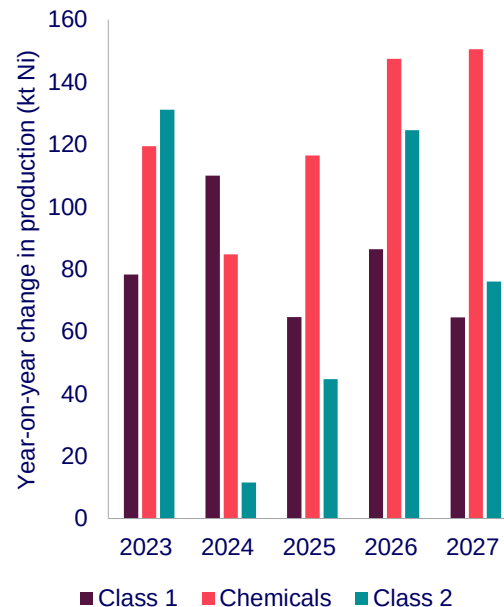
There is potential for swing production between sulphate and metal depending on relative prices and demand for sulphate.

Growth in NPI/FeNi output has fallen significantly due to operational closures, project delays and anticipated slower new furnace line additions in Indonesia. Output is forecast to rise by 11% by 2027. Overall, forecast Class II production for 2025 is flat on 2024.

Refined production by type



Growth in refined production by class



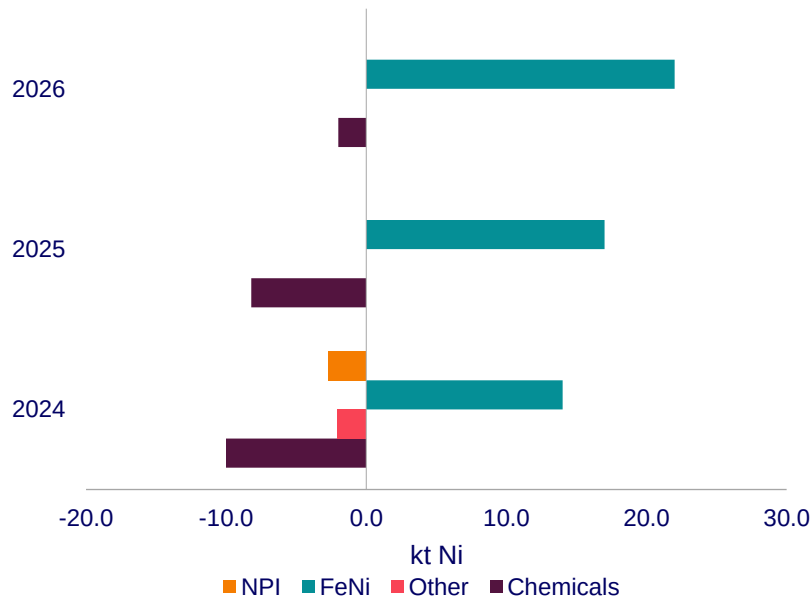
Changes in forecast nickel production

Increased FeNi in Korea and the resumption of production in Myanmar

Our January forecasts for 2025 and 2026 are increased from those of December 2024 by 9 kt and 20 kt, respectively.

The increases are due to the apparent resumption of operations at Tagaung Taung in Myanmar. Production was previously halted due to fighting between government and resistance forces. After re-starting, Tagaung is expected to produce near to capacity in 2025.

Main changes in finished nickel output



Source: Wood Mackenzie

Matte and MHP payability forecasts

Intermediate payabilities should remain largely flat over the forecast period

Both matte and MHP payabilities were flat in December over November, averaging 85% and 82%, respectively. Both then fell in January to 84% and 80%, respectively. Our view is that payabilities will remain relatively flat over the forecast period, although averaging slightly lower than 2024 for MHP, and slightly higher for matte.

	2024	2025	2026	2027
Matte	82%	83%	83%	83%
MHP	80%	79%	79%	79%

Source: Wood Mackenzie

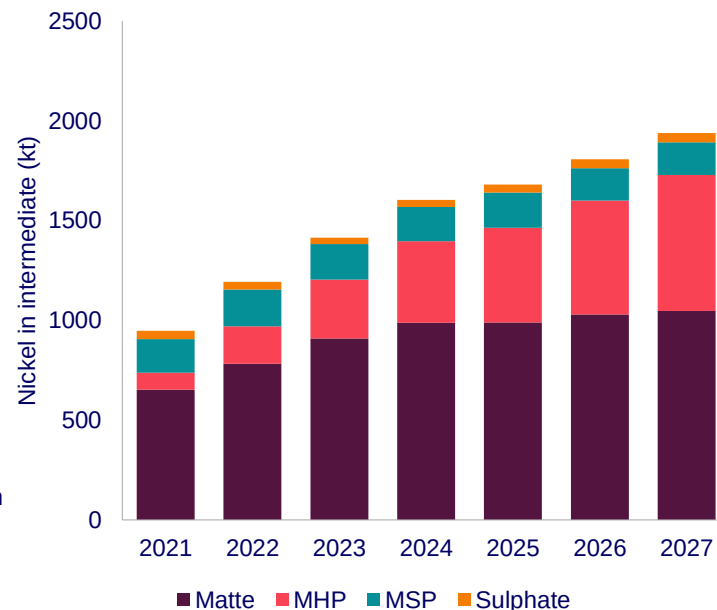
Payabilities are calculated based on current metal and sulphate prices, and conversions costs – providing an effective payability that can sustain a margin at a given price.

By 2026, nickel-in-matte production will increase by 6% over 2024 levels.

Nickel-in-MHP output will increase by 67% over the same time frame, solely through capacity additions in Indonesia.

There is considerable potential for the output of both MHP and matte to increase more strongly than shown, almost entirely from increases in Indonesian production.

Nickel-in-intermediates production



Source: Wood Mackenzie

Nickel sulphate production

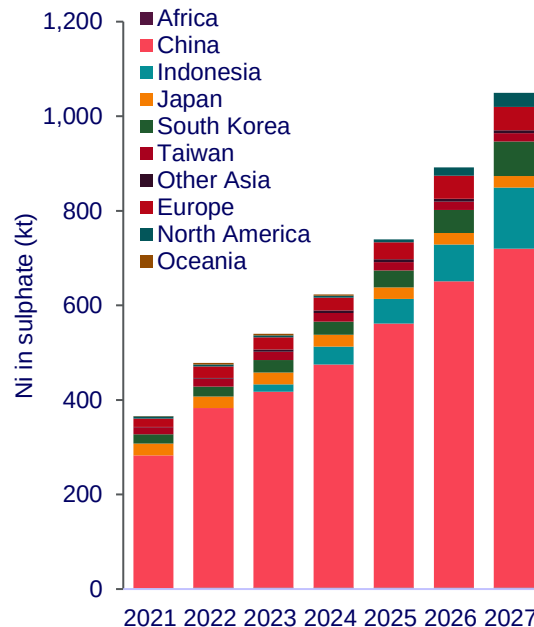
Nickel-in-chemicals production (predominantly as sulphate) will increase by 19% a year through to 2027

China currently accounts for 76% of global nickel sulphate production. This global position will decline to 68% by 2027 on the assumption that new plants in Indonesia (e.g. Excelsior, which will have optionality for metal or MHP) produce sulphate.

Metal dissolution in China is now at minimal levels following the substantial increase in intermediates production in Indonesia.

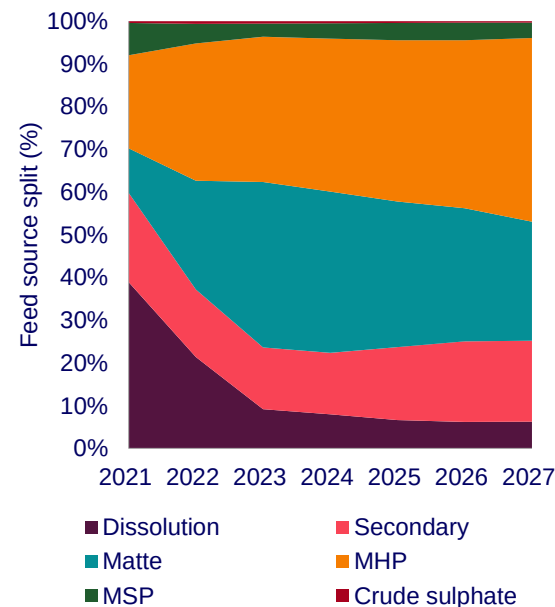
Potential exists to increase chemicals output should demand require it. But at present substantial tonnages of sulphate are being converted to metal.

Global nickel sulphate production



Source: Wood Mackenzie

Raw material feeds for NiSO₄



Source: Wood Mackenzie

Nickel supply - Indonesia

Indonesia increases permitted mining tonnages for 2025

After much speculation in late 2024 and in the early part of 2025, there appears to have been a “definitive” announcement on how much Indonesian nickel ore is permitted for mining in 2025. On 23 January 2025, the Indonesian Nickel Mining Association (APNI) announced at a parliamentary hearing that Indonesia had approved a nickel mining quota of 298.5 Mwmt, up from 271.89 Mwmt in 2024. This was in stark contrast to previous indications that quotas would be cut in 2025, some even suggesting figures as low as 150 Mwmt.

Our view was that a restriction on ore tonnage was almost certainly coming, but not to the extent of a cut to 150 Mwmt. That anticipated cut in quotas, for now, has not been applied in 2025. What remains difficult to assess is what ore has been permitted. More RKEF lines as well as HPAL projects are expected to come on stream in 2025. On a purely pro-rata basis – assuming all the ore permitted for 2024 was processed - the 2025 quota is sufficient to support an additional 200 kt contained nickel production. We presently show an increase in NPI and intermediates of 150 kt nickel.

It may be that the new quota will ease any perceived ore tightness in Indonesia, reducing the need to import ores from the Philippines and lower ore prices closer to the benchmark HPM price, thereby cutting the premiums that had been applied since early in 2024.

The announced quota will support a production increase and probably reduce domestic ore purchase prices. However, it will not “control” the nickel price in the way Indonesia had indicated it wished to do – to “manage” the nickel price at a higher level than it currently trades.

The change of direction from almost certain cuts in ore output to an increase may well have been influenced by the Trump administration rolling back EV mandates. This policy reversal means that the prospect of US investment in Indonesia is less likely. It is just possible that this had an influence on supporting revenue and growth to the nickel industry.

Nickel supply - Indonesia

Filipino ore exports may increase in 2025

Following delays in the issuance of RKAB licences, ore availability in Indonesia last year became restricted and producers turned to the Philippines as a source of alternative supply.

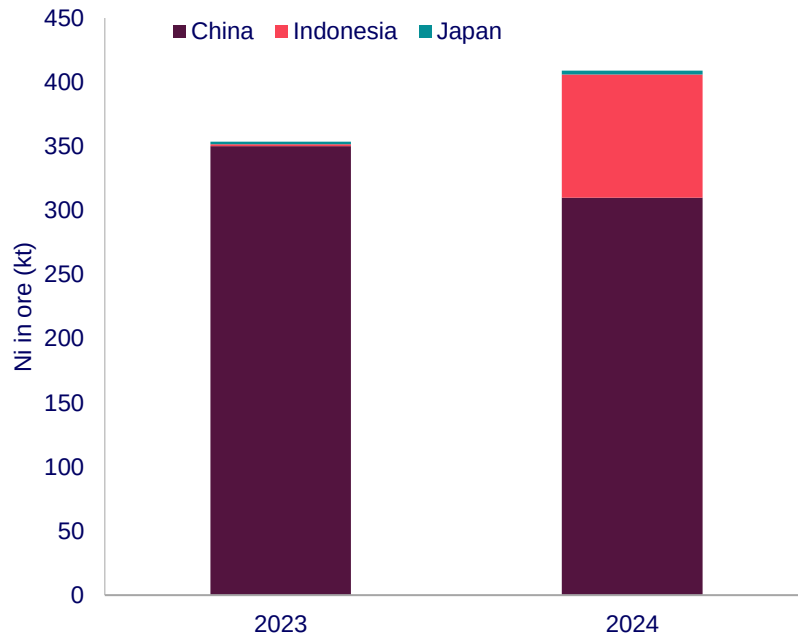
In 2024, around 95 kt nickel-in-ore was imported by Indonesia from the Philippines, predominantly mid-grade ore grading 1.3-1.5% Ni.

For 2025, it remains unclear exactly what Indonesian mine production will be. However, the apparent “plan” is to restrict rather than encourage growth in mine output.

While the same can be said of the Philippines in terms of forecast production levels, the country is also reforming its mining laws to boost the nickel industry. Iipilan Nickel Corp. (an operating subsidiary of Global Ferronickel Holdings Inc.) as well as DMCI Mining both have expansions plans to more than double ore production (from ca. 2.5 Mwmt a year combined to 6 Mwmt a year).

Growth in Filipino nickel mine output appears more likely than it has for some time.

Filipino nickel-in-ore exports by destination in 2024



Source: Global Trade Tracker; wood Mackenzie

Nickel supply - China

Xinhai raises NPI grade with third-party purchases

We have revised down our 2024 estimate for Shandong Xinhai's nickel-in-NPI production in China due to apparent double counting. We noted in May 2024 that monthly production levels had risen quite sharply. This appeared at odds with a decline in April and the shift in ore availability at that time.

Xinhai had, we believe, been taking ores from New Caledonia as part of its raw materials feed mix at its facilities in China. The decline in domestic output ties in with the civil unrest that erupted in New Caledonia in May.

Xinhai began to procure FeNi/NPI from Indonesia at that time, apparently to increase the grade of nickel in its NPI product. The loss of higher-grade ores from New Caledonia would certainly have resulted in a decline in nickel production, and probably NPI grade. While we cannot be positive that purchasing third-party FeNi and NPI were a direct result of the New Caledonian unrest, it would appear to be a very likely cause.

In addition, there must be a clear arbitrage between Indonesian and Chinese NPI in order to make these transactions economically viable. This will be a driving factor as to whether this practice continues in 2025, as will New Caledonian ore exports.

Overall, in 2024, the three Xinhai facilities in Shandong combined “produced” 130 kt nickel-in-NPI, of which almost 25% was purchased. Purchases averaged more than 4 kt nickel a month over the second half of 2024.

We are investigating to determine if any other NPI facilities in China have undertaken a similar strategy to ensure we are not double counting other nickel units.

Nickel supply analysis

Ramu outage supports MHP payabilities. The Ramu HPAL plant in Papua New Guinea has suspended production in part of the plant due to an acid plant failure. The failure of a blower will result in an estimated cut in output of 6-10 kt nickel in MHP, according to the company. Nickel production for 2025 was slated at around 35 kt before the acid plant failure.

According to some sources, MHP payabilities were expected to decline in Q1 2025 (at least after the Lunar Festival) due to poor battery sector demand. However, the loss of Ramu material in the near-term will support current payable levels (ca. 79.5%, having fallen two percentage points month-on-month). In part, this is because Ramu's partner, MCC, had recently doubled capacity (to 50 ktpa nickel-in-sulphate) at its Caofeidian refinery in Hebei. Caofeidian has the offtake for a large portion of Ramu's MHP, and the loss of feed has forced the company into securing larger tonnages in the third-party market, which has helped to maintain current payable levels.

While the second acid plant remains in operation it is contemporaneous with the one that failed. It is therefore possible that despite an overhaul in September 2024, the second acid plant may also need to be taken offline at some point for a refit.

Vale undertaking strategic review of Thompson operations As part of a broader optimisation process across Vale Base Metals (VBM) assets, VBM has launched a strategic review to “explore and evaluate a range of alternatives, including the potential sale, for its mining and exploration assets in Thompson, Manitoba.”

Production in Thompson began in 1961. Currently, only the Thompson mine remains, producing around 10 ktpa nickel in concentrate. Following the closure of the Thompson smelter and refinery in the second half of 2018, nickel concentrate has been shipped for processing at Sudbury and/or Long-Harbour. VBM no longer reports any reserves for Thompson (stating in its 20-F filing “We continue to operate and produce from our mine in Thompson, and we continue to seek economic viability to convert mineral resource into mineral reserves.”). However, measured and indicated mineral resources were 19.1 Mt grading 1.98% Ni, with an additional inferred resource of 6.8 Mt grading 1.9% Ni, as of December 2023. At current ore mining rates of around 700 ktpa, this would imply a mine life of 35 plus years.

The key point to consider is “we seek economic viability”. Wood Mackenzie's analysis indicates cash costs at Thompson are over US\$22,000/t nickel.

The broader optimisation process is expected to conclude in the second half of 2025. Considering our forecast for nickel prices and Thompson's current economic condition, it seems improbable that a buyer could be found to maintain operations. A shift to care and maintenance is far more likely. However, this would result in the loss of much-needed concentrate for the Sudbury and Long Harbour facilities during a period of tight concentrate supply, and this wider economic impact across the integrated facilities could be its lifeline, at least in the near term.

Appendix

Key forecasts

Nickel key forecasts

	2024		2025		2026		2027	
Global (kt)								
Mine production	3906	+2.4%	3993	+2.2% ▲	4367	+9.3% ▲	4692	+7.5%
Smelter production	3675	+5.7%	3724	+1.3% ▲	4024	+8.1% ▲	4319	+7.3%
Refined production	3662	+6.0%	3830	+4.6% ▲	4190	+9.4% ▲	4502	+7.4%
Consumption	3378	+6.8%	3732	+10.5% ▲	3939	+5.5% ▲	4188	+6.3%
Market balance	284		98		251		314	
Cash LME price								
(US\$/t - current)	16813		15788		15460		15065	
(c/lb - current)	763		716		701		683	

Source: Wood Mackenzie

Nickel sulphate key forecasts

	2024		2025		2026		2027	
Global (kt Ni)								
NiSO4 production	623	+15.5%	757	+21.5% ▲	915	+20.8% ▲	1091	+19.3%
Consumption in EV	520	+5.8%	641	+23.1% ▲	751	+17.2% ▲	887	+18.2%
Consumption in plating	39	+3.6%	40	+2.3% ▲	41	+2.3% ▲	42	+2.2%
Consumption in other	17	+7.5%	18	+6.1% ▲	18	+1.6% ▲	18	+1.2%
Total NiSO4 consumption	576 ▲	+5.7%	698 ▲	+21.2%	810 ▲	+16.0%	947	+17.0%
Market balance	47		59		105		144	
NiSO4 price CHINA								
(US\$/t - current)	3441		3232	-6.1% ▲	3094	-4.3% ▲	3013	-2.6%
(US\$/t Ni - current)	15498		14560 ▲	-6.1%	13936 ▲	-4.3%	13574	-2.6%
Premium/discount to LME								
(US\$/t Ni = current)	-1315		-1228 ▲	-6.6%	-1524 ▲	+24.1%	-1491	-2.2%
NiSO4 price ROW								
(US\$/t - current)	4161		3907	-6.1% ▲	3834	-1.9% ▲	3744	-2.4%
(US\$/t Ni - current)	18745		17601 ▲	-6.1%	17272 ▲	-1.9%	16864	-2.4%
Premium/discount to LME								
(US\$/t Ni = current)	1932		1813 ▲	-6.2%	1812 ▲	-0.0%	1799	-0.8%

Source: Wood Mackenzie; the Argus Media group

Quarterly nickel consumption (kt)

Quarterly global nickel consumption in stainless steel

	2024 Year	2025					2026				
		Q1	Q2	Q3	Q4	Year	Q1	Q2	Q3	Q4	Year
SS output	62891	15556	16722	17000	17101	66499	16686	17814	17780	17737	70017
Aust output	50583	12453	13420	13699	13787	53359	13393	14317	14253	14254	56217
Scrap (%)	33	31	31	31	30	31	32	32	31	32	32
Aust (%)	80	80	80	81	81	80	80	80	80	80	80
Primary Ni	2183	558	597	614	618	2388	592	625	627	624	2468

Source: Wood Mackenzie

Quarterly refined nickel consumption by first use

	2024 Year	2025					2026				
		Q1	Q2	Q3	Q4	Year	Q1	Q2	Q3	Q4	Year
TOTAL	3378	872	925	957	979	3732	949	987	997	1006	3939
Stainless steel	2183	558	597	614	618	2388	592	625	627	624	2468
Non-stainless	1195	313	328	343	360	1344	358	362	370	382	1472
Alloy steel	94	25	25	24	25	98	25	25	24	25	100
Non-ferrous alloys	252	66	66	66	67	266	69	68	68	69	275
Plating	177	45	45	45	46	182	46	46	46	46	186
Battery precursors	520	137	152	169	182	641	177	182	191	200	751
Foundry	69	18	18	17	18	70	18	18	17	18	71
Other	83	22	22	22	22	88	22	22	22	22	89

Source: Wood Mackenzie

Nickel stocks and premiums

	2024	2024	2024	2024	2024	2024	2024	2024	2025
		Q3	Q4	Aug	Sep	Oct	Nov	Dec	Jan
Stocks - kt									
Antwerp	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Baltimore	0.3	0.4	0.3	0.5	0.4	0.4	0.3	0.3	0.2
Busan	29.8	25.1	29.8	23.1	25.1	25.9	29.0	29.8	33.1
Detroit	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Dubai	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Gwangyang	8.7	9.7	8.7	9.8	9.7	9.7	9.6	8.7	9.3
Johor	9.1	5.2	9.1	5.2	5.2	8.7	11.7	9.1	8.9
Kaohsiung	38.9	25.4	38.9	21.8	25.4	34.0	39.0	38.9	42.8
Liverpool	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Mobile	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Port Klang	11.6	9.8	11.6	9.8	9.8	12.7	12.5	11.6	11.0
Rotterdam	32.1	30.9	32.1	26.1	30.9	30.8	31.0	32.1	32.4
Singapore (City)	30.0	22.7	30.0	21.0	22.7	22.5	28.1	30.0	32.6
Missingen	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total LME stocks	162.5	131.2	162.5	119.3	131.2	146.8	163.1	162.5	172.3
Cancelled warrants	11.8	8.1	11.8	8.9	8.1	8.1	11.3	11.8	10.1
Shanghai Futures Exchange	36.1	25.5	36.1	22.6	25.5	28.0	33.0	36.1	36.0
Prices									
LME									
Cash (US\$/t)	16807	16255	16034	16246	16113	16800	15737	15466	15400
3-month (US\$/t)	17057	16512	16289	16488	16359	17064	16000	15701	15622
Spread (3-month, cash)	250	257	255	242	246	264	263	235	222

	2024	2024	2024	2024	2024	2024	2024	2024	2025
		Q3	Q4	Aug	Sep	Oct	Nov	Dec	Jan
SHFE									
Front month settlement RMB/t	136179	129274	126651	129686	125381	129622	125750	125309	125540
3-month settlement RMB/t	136913	129922	127380	130301	126052	130535	126403	125979	125987
Front month settlement US\$/t	16746	15961	15584	16045	15682	16150	15446	15239	15200
3-month settlement US\$/t	16836	16041	15674	16121	15766	16264	15526	15321	15254
Physical premiums (US\$/t)									
Europe plating	401	378	327	379	369	344	314	324	313
Europe melting	277	300	268	292	294	313	247	244	238
Europe FPC	209	197	156	194	193	170	146	152	139
Exchange Rates/US\$									
Australian \$	1.513	1.493	1.526	1.502	1.478	1.491	1.530	1.570	1.607
Canadian \$	1.367	1.364	1.394	1.365	1.355	1.373	1.397	1.418	1.438
Yen	151.44	149.22	151.87	146.23	143.22	149.63	154.04	152.52	156.86
Yuan	7.197	7.168	7.192	7.153	7.075	7.103	7.205	7.277	7.309
Euro	0.754	0.910	0.935	0.908	0.900	0.917	0.940	0.953	0.966

Source: LME, SHFE, IMF, the Argus Media Group, Wood Mackenzie

Contacts

Research Director – Nickel Markets

Sean Mulshaw (Guildford, UK)

T +44 1483 596637

E sean.mulshaw@woodmac.com

Principal Analyst – Nickel Markets

Adrian Gardner (Palm Coast, FL, USA)

T +1 832 882 8024

E adrian.gardner@woodmac.com

Research Director - Nickel

Andrew Mitchell (Guildford, UK)

T +44 1483 596610

E andrew.mitchell@woodmac.com

Senior Consultant - Metals

Yingchi Yang (Shanghai, China)

M +86 180 1540 1570

E Yingchi.Yang@woodmac.com

Research Associate - Metals

Panav Mohindra (New Delhi, India)

T +91 7042 534353

E Panav.mohindra@woodmac.com

Wood Mackenzie Client Helpdesk

E support@woodmac.com

Americas +1 713 470 1700

Asia Pacific +65 6518 0888

Europe +44 131 243 4477

Disclaimer

These materials, including any updates to them, are published by and remain subject to the copyright of the Wood Mackenzie group ("Wood Mackenzie"), or its third-party licensors ("Licensors") as relevant, and are made available to clients of Wood Mackenzie under terms agreed between Wood Mackenzie and those clients. The use of these materials is governed by the terms and conditions of the agreement under which they were provided. The content and conclusions contained are confidential and may not be disclosed to any other person without Wood Mackenzie's prior written permission. Wood Mackenzie makes no warranty or representation about the accuracy or completeness of the information and data contained in these materials, which are provided 'as is'. The opinions expressed in these materials are those of Wood Mackenzie, and do not necessarily represent our Licensors' position or views. Nothing contained in them constitutes an offer to buy or to sell securities, or investment advice. Wood Mackenzie's products do not provide a comprehensive analysis of the financial position or prospects of any company or entity and nothing in any such product should be taken as comment regarding the value of the securities of any entity. If, notwithstanding the foregoing, you or any other person relies upon these materials in any way, Wood Mackenzie does not accept, and hereby disclaims to the extent permitted by law, all liability for any loss and damage suffered arising in connection with such reliance.

Copyright © 2024, Wood Mackenzie Limited. All rights reserved.



Europe	+44 131 243 4477
Americas	+1 713 470 1700
Asia Pacific	+65 6518 0888
Email	contactus@woodmac.com
Website	www.woodmac.com

Wood Mackenzie™ is a trusted intelligence provider, empowering decision-makers with unique insight on the world's natural resources. We are a leading research and consultancy business for the global energy, power and renewables, subsurface, chemicals, and metals and mining industries.

For more information visit: woodmac.com

WOOD MACKENZIE is a trademark of Wood Mackenzie Limited and is the subject of trademark registrations and/or applications in the European Community, the USA and other countries around the world.